

Experiment No: 9

Title:

Implementation of `localStorage` and `sessionStorage` to Save User Preferences such as Theme Settings.

Software/Tools Required:

- Visual Studio Code
- Web Browser such as Google Chrome
- HTML, CSS, and JavaScript

Theory:

Modern web browsers provide the **Web Storage API**, which enables web applications to store data locally on the user's device without requiring a database or server. The Web Storage API consists of two storage mechanisms: **`localStorage`** and **`sessionStorage`**. Both store data as key-value pairs and allow JavaScript to save, retrieve, update, and remove information efficiently.

`localStorage` stores data permanently until it is explicitly removed by the user or the application. The stored data remains available even after the browser is closed and reopened, making it suitable for saving long-term user preferences such as theme settings, language preferences, or recently visited pages.

`sessionStorage`, on the other hand, stores data only for the duration of the current browser session. The stored information is automatically cleared when the browser tab or window is closed. It is useful for storing temporary data such as login session details, form data, or temporary user selections.

JavaScript provides methods such as `setItem()`, `getItem()`, `removeItem()`, and `clear()` to manage data in both storage types. Since the data is stored as strings, complex objects can be converted into JSON format using `JSON.stringify()` before storage and restored using `JSON.parse()` when retrieved.

This experiment demonstrates how client-side storage can be used to enhance user experience by remembering preferences, reducing repetitive input, and providing personalized interactions without relying on server-side storage.

Result/Conclusion:

Thus, `localStorage` and `sessionStorage` were successfully implemented to store user preferences and temporary session data. The application effectively saved and retrieved theme settings, demonstrating the practical use of the Web Storage API for creating personalized and user-friendly web applications.

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Seminar Schedule Planner

Click on a seminar to view and save its details.

Day	Begin	End	Topic
Monday	8:00 a.m.	5:00 p.m.	Introduction to XML
Monday	8:00 a.m.	11:00 a.m.	Validity: DTD and Relax NG Saved
Tuesday	11:00 a.m.	2:00 p.m.	XPath Day Saved
Tuesday	2:00 p.m.	5:00 p.m.	XSL Transformations Saved
Wednesday	8:00 a.m.	12:00 p.m.	XSL Formatting Objects

Save Topic Save Day Clear All

Tuesday's complete schedule saved to sessionStorage.

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Username

Mohammed

Saved permanently with localStorage.

Choose Theme

Dark

Choose Accent Color

Amber

Save Preferences Clear Preferences

LOCALSTORAGE

No preferences saved

Persists after closing and reopening the browser.

SESSIONSTORAGE

Theme: dark

Used for the current page/session state.